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Data Cleaning & Preparation

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Project:
Sales Dataset Data Cleaning using Python & Pandas

Technology Used:
Python, Pandas, NumPy, Jupyter Notebook

Introduction

This project focuses on cleaning and preparing a raw sales dataset using Python and Pandas. The main objective of the project is to improve data quality by handling missing values, removing duplicate records, and correcting inconsistent data formats. Data cleaning is an important step in data analytics because accurate and clean data helps generate reliable insights.

Objective

Objectives of the project:

- Identify missing values
- Remove duplicate records
- Correct date and text formatting
- Prepare a clean dataset for analysis

Tools & Technologies

- Python
- Pandas
- NumPy
- Jupyter Notebook
- Excel

Dataset Information

The dataset contains sales-related information with multiple columns such as customer details, product information, order dates, payment methods, and coupon codes. The dataset contained missing values and required preprocessing before analysis.

Steps Performed

1. Loaded dataset using Pandas
2. Checked dataset structure and missing values

3. Removed duplicate rows
4. Filled missing CouponCode values
5. Standardized date formats
6. Removed extra spaces from text columns
7. Verified cleaned dataset
8. Saved cleaned dataset into a new Excel file

Results

After performing data cleaning:

- Missing values were handled successfully
- Duplicate records were removed
- Date formats were standardized
- Dataset became clean and analysis-ready

Conclusion

This project helped in understanding the importance of data cleaning and preparation in data analytics. Using Python and Pandas, the dataset was successfully cleaned and transformed into a reliable format for future analysis and visualization.